

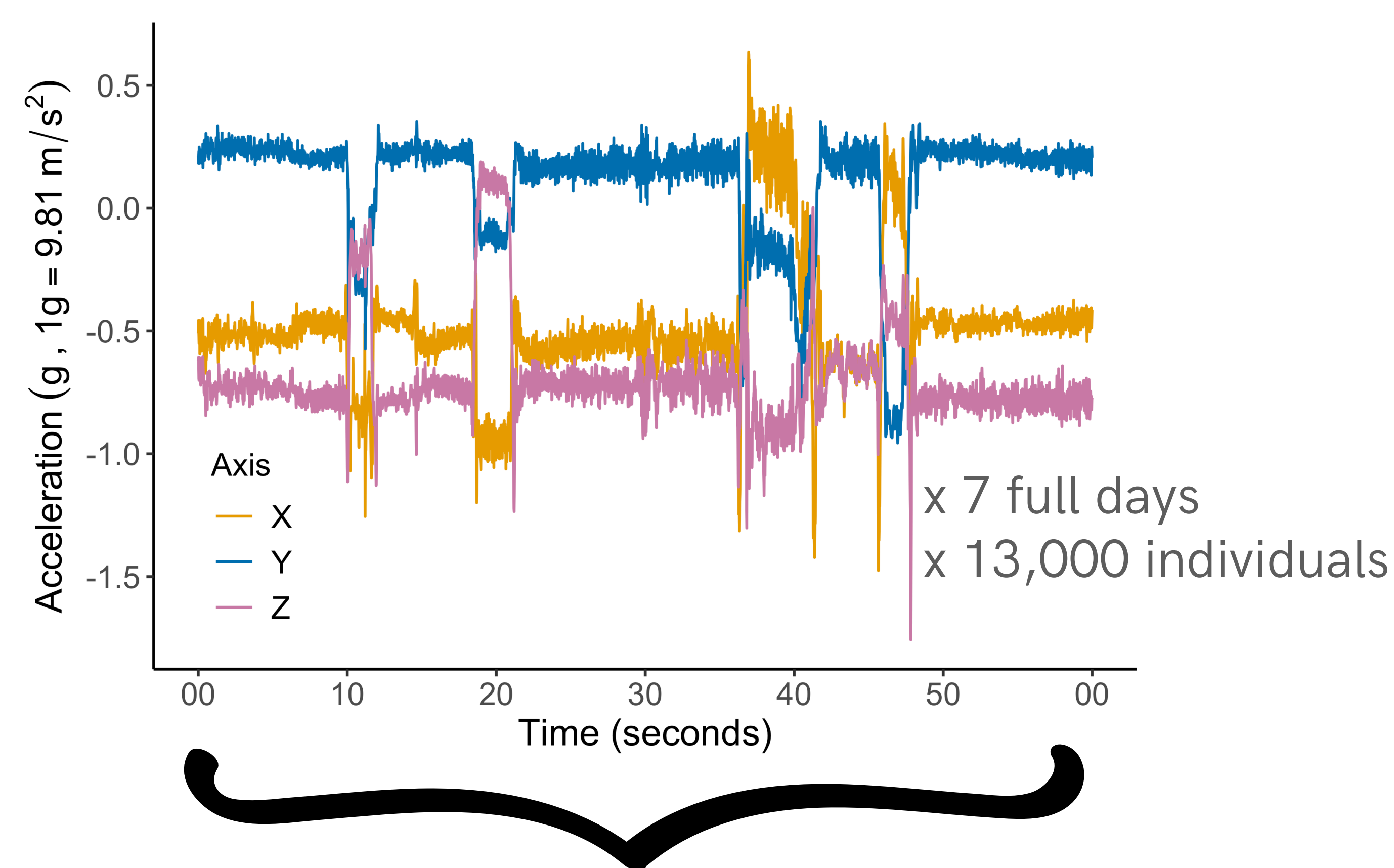
FINGERPRINTING WALKING in a large epidemiological study



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DATA

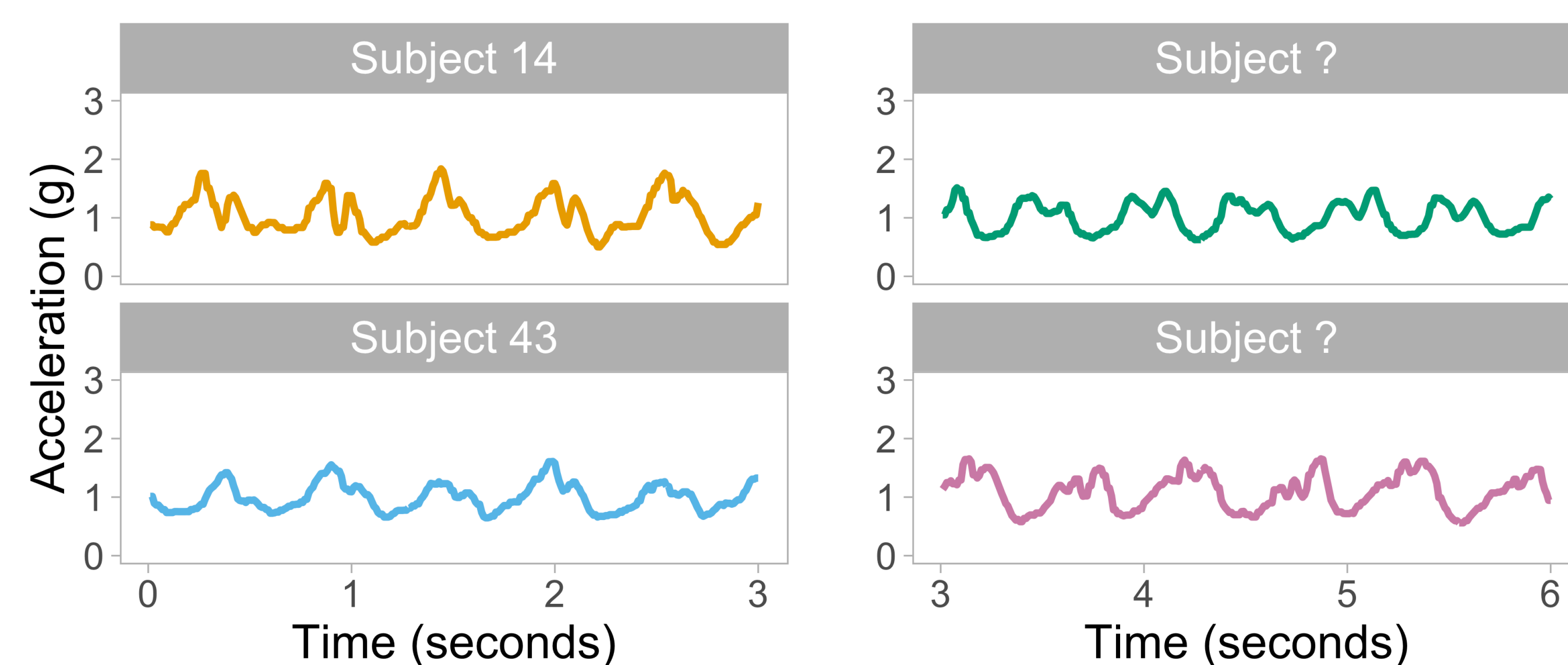
- Ntl Health & Nutrition Examination Survey (NHANES)
- Questionnaires and laboratory tests on large, **nationally representative** sample of Americans
- 2011-2014: survey included accelerometry sub-study



1 minute of data
80 observations per second per dimension

OBJECTIVE

Leverage accelerometry-derived walking patterns for biometric identification

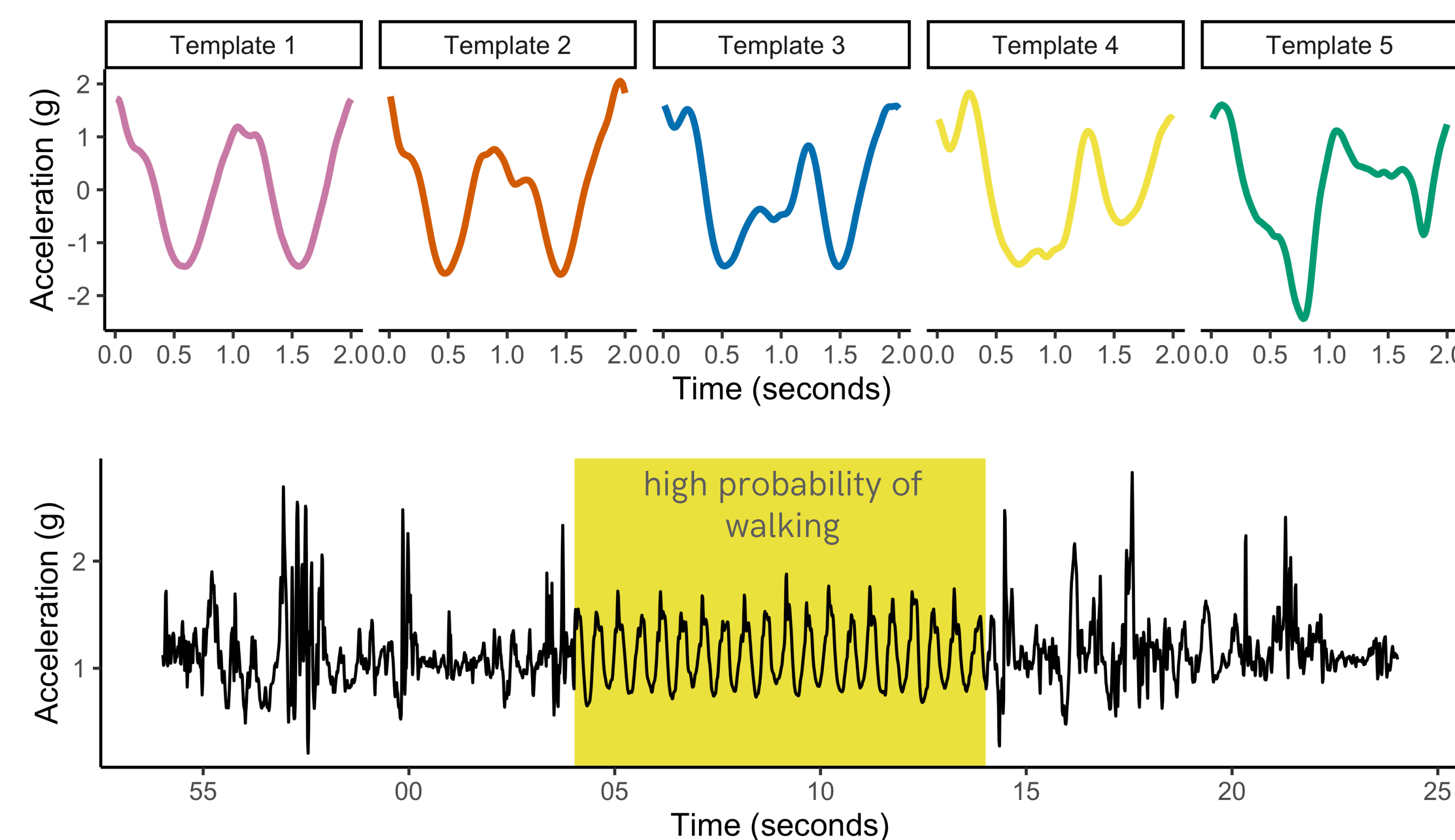


METHODS

1. Identify time periods with high probability of being walking
2. Derive **grid-cell predictors** for identification models
3. Implement various identification models under different train/test scenarios

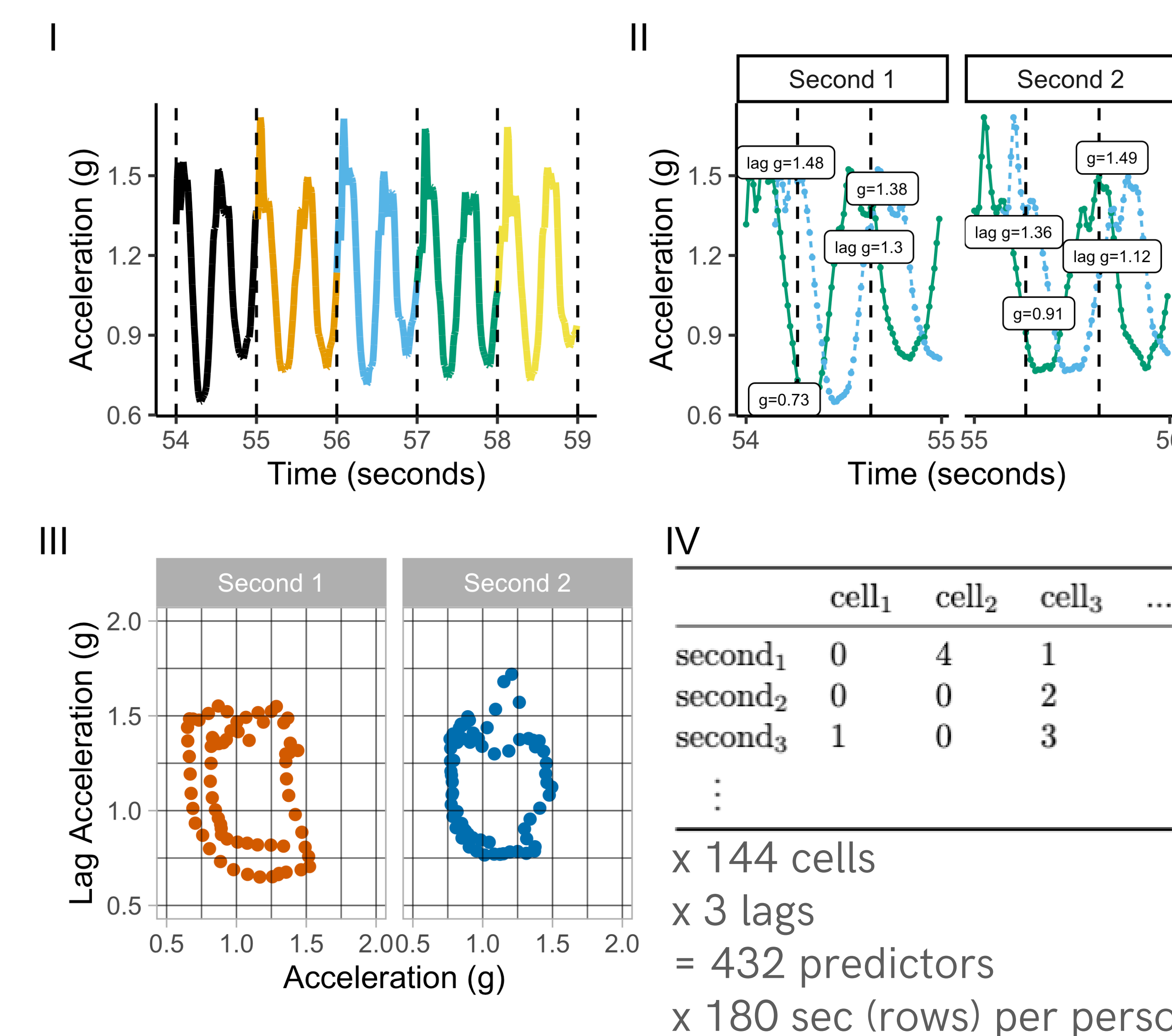
1. IDENTIFY WALKING

Use Adaptive Empirical Pattern Matching (ADEPT)¹ to identify walking periods from raw accelerometry data



2. DERIVE PREDICTORS

- I. Break walking periods into 1-second segments
- II. Compute empirical joint distribution of acceleration and lag acceleration at each 1/80th of a second for three lags
- III. Tabulate number of acceleration, lag acceleration pairs in each 0.25x0.25g square on the [0,3]gx[0,3]g grid
- IV. Create predictor matrix

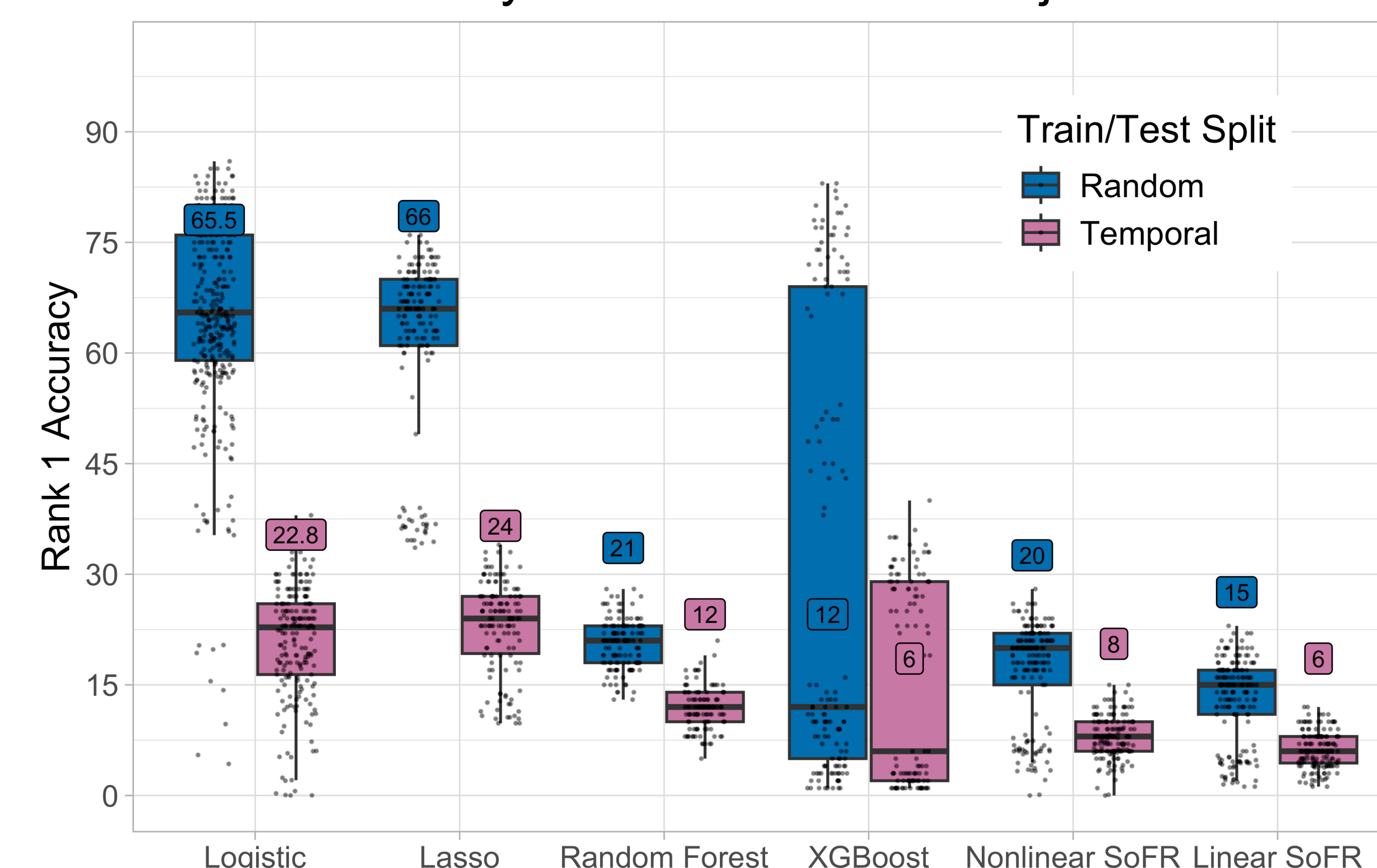


3. FIT MODELS

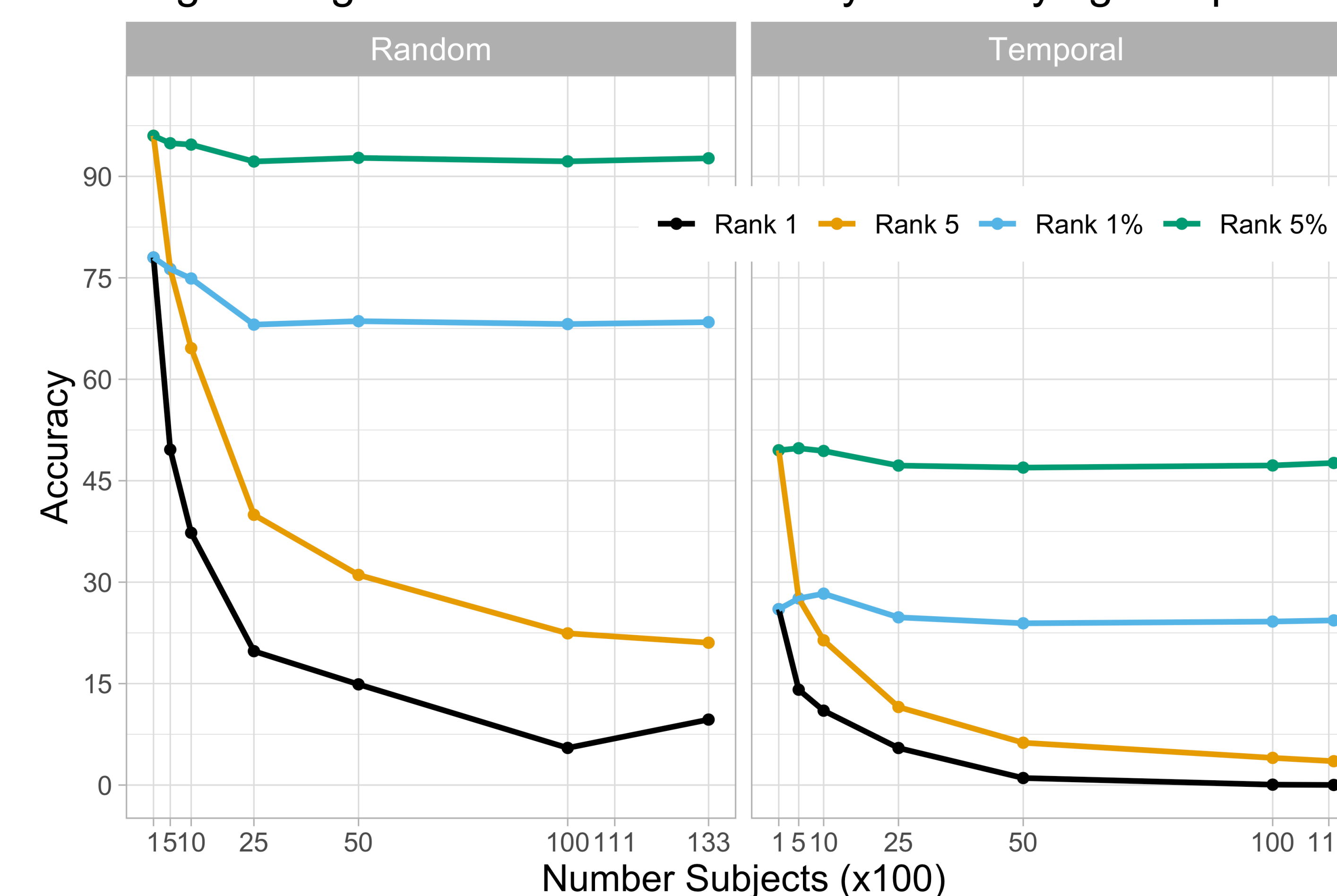
- One vs. rest classification paradigm with varying size subsets of the entire sample
- Logistic regression, machine learning (random forest, XGBoost), scalar on function regression
- Training and testing **randomly sampled**
- Training from days 1-3, testing from days 4-7 (**temporal**)

RESULTS

Rank 1 accuracy in subsets of 100 subjects



Logistic regression models: accuracy with varying sample size



TAKEHOME

- Accelerometry data can identify individuals from walking
- Implications for epidemiology, health, privacy/security
- Scalable to large datasets

